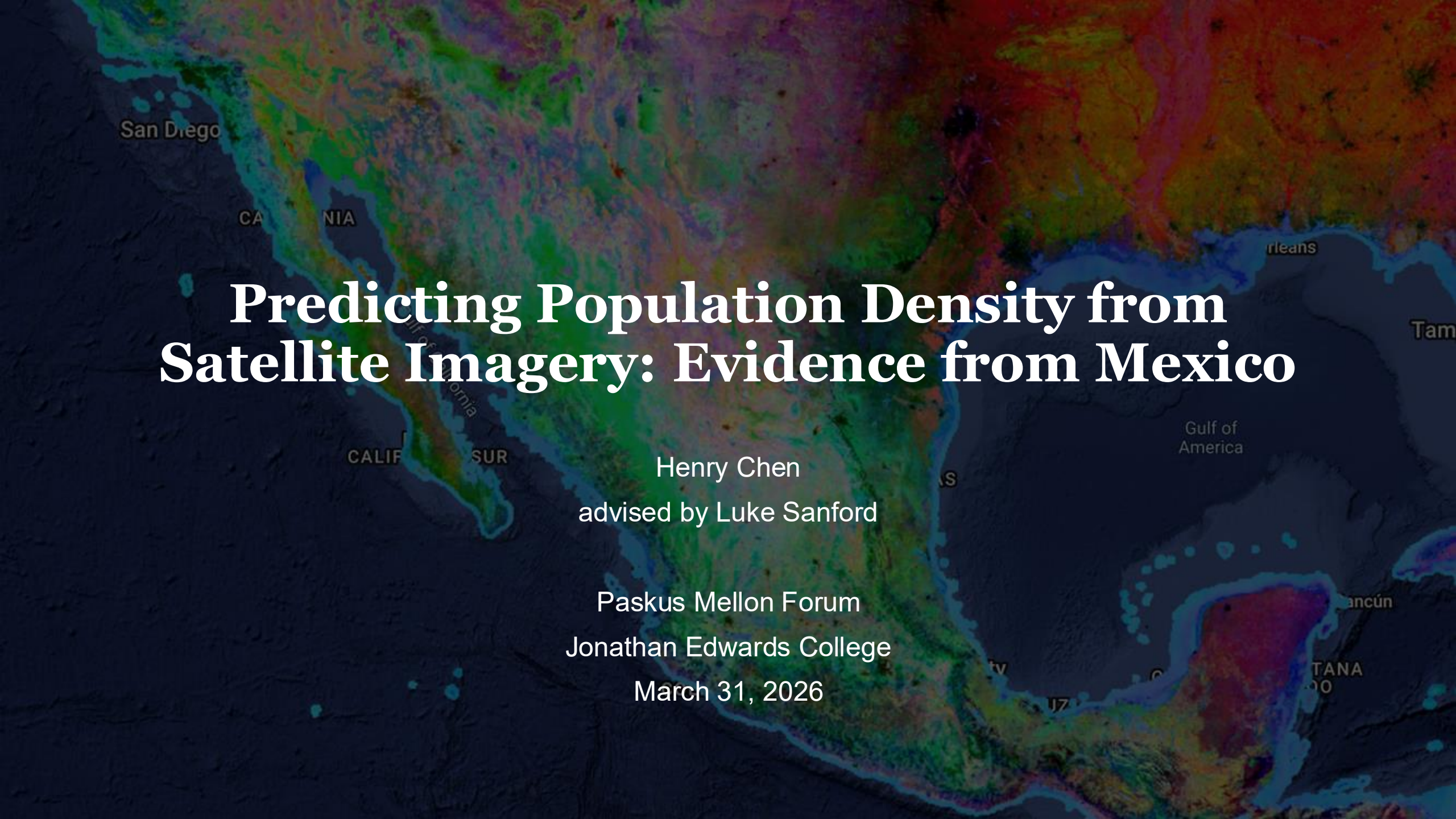




Predicting Population Density from Satellite Imagery: Evidence from Mexico

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Paskus Mellon Forum

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Agenda

- **Motivation**
- Intervention
- Methodology
- Results

Population data underlies many critical decisions in public life...



Disaster response

Evacuation, aid distribution



Infrastructure

Transit, utilities, housing



Epidemic modeling

Spread forecasting, vaccination



Development economics

Aid targeting, growth modeling



Electoral redistricting

District boundaries, representation



Crime & policing

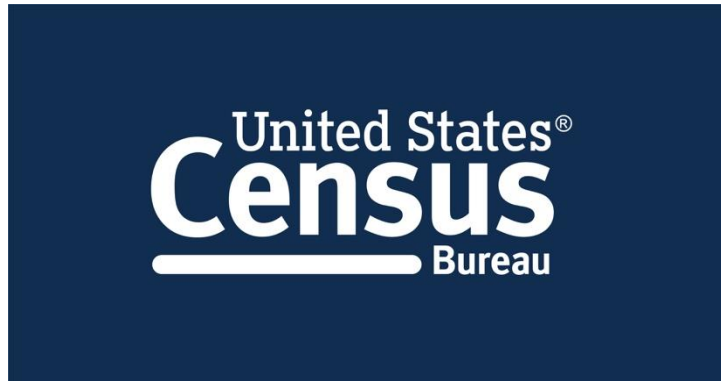
Resource deployment, risk modeling

\$2.8T+

in federal spending allocated annually using census-derived formulas¹

1. US Census Bureau press release CB23-102, June 14, 2023.

... yet population data is often not frequently collected or updated.



Latency problem: Robust census data only collected **every 10 years**

Granularity problem: Between censuses, there is often **no high-frequency full count** at the neighborhood/block levels

This means researchers often rely on incomplete **data up to a decade old**

Motivating question:

How can we address temporal sparsity and spatial granularity to make population data more accurate?

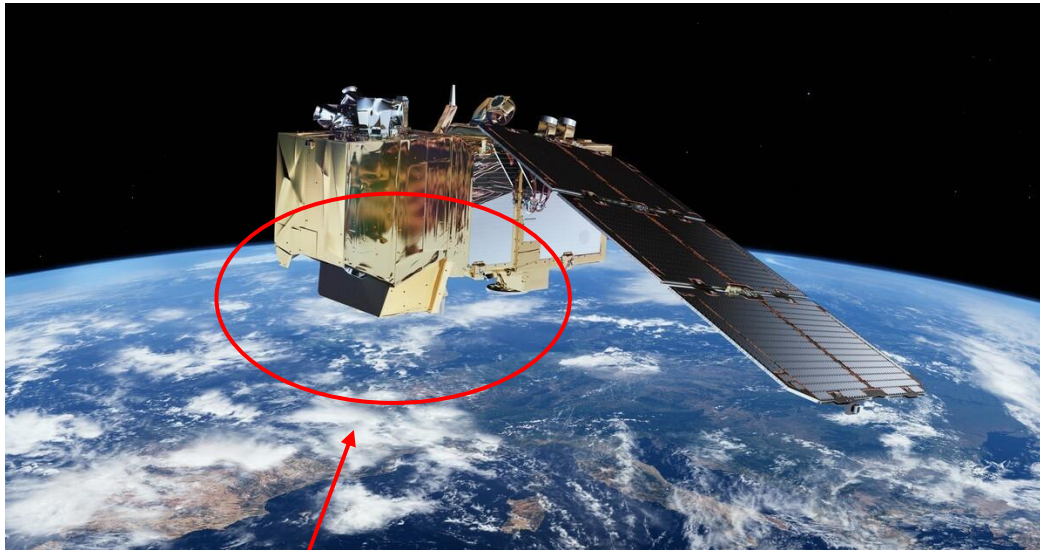
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Satellite imagery can solve both problems

It's **global**, **continuously updated**, available at **fine resolutions**

Sentinel-2 satellite (launched in 2015)



camera + sensors capture high resolution photos and infrared, radar, thermal, multispectral signals of Earth



Census survey data vs. satellite imagery

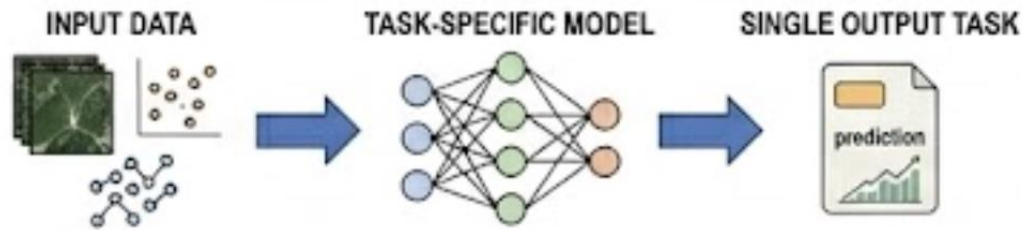
	CENSUS	SATELLITE
Latency	10 years update cycle	1 day update cycle
Granularity	tract finest resolution	10 m pixel resolution
Coverage	varies by country	global including data-sparse regions ✨

Research question:

Does satellite imagery contain enough signal about human settlement patterns to support accurate population estimates?

Previous studies of this question predict population density using task-specific trained models

How does a task-specific trained model work?



Results in the literature

Using geospatial data, traditional task-specific models capture **60–75%** of the variation in population density in the real world.¹

Limitations

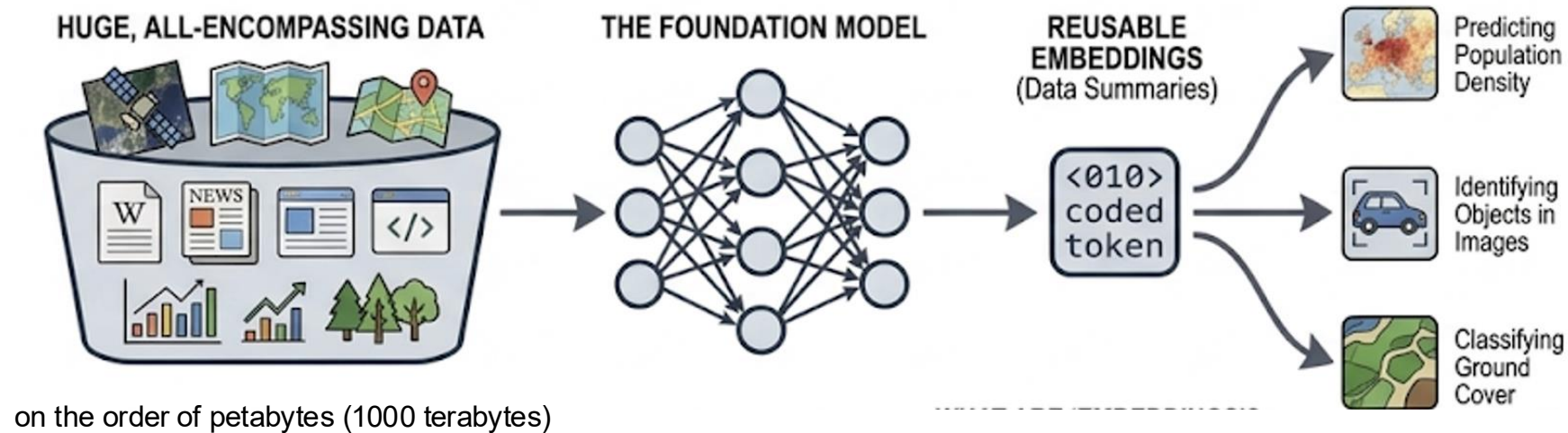
Doesn't scale to new regions / doesn't generalize

Requires expensive re-training of model for each new task / geography

Limited by the data it was trained on

1. Stevens et al. (2015), WorldPop project, various studies on deep learning (CNN) approaches to the geospatial problems

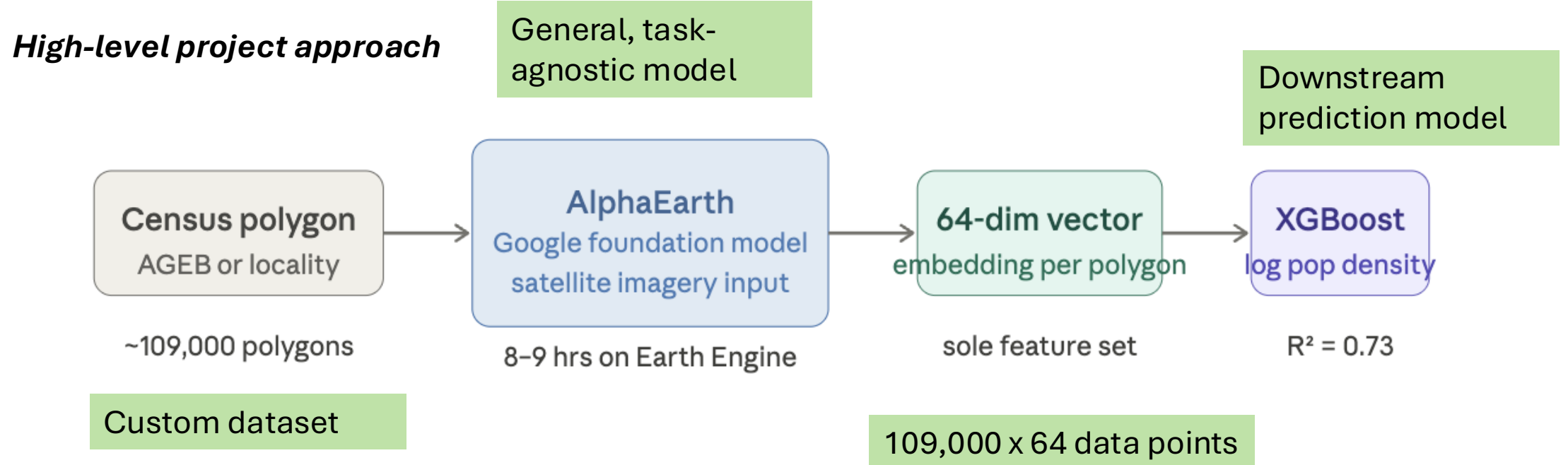
My project asks whether task-agnostic models can perform just as well.



Agenda

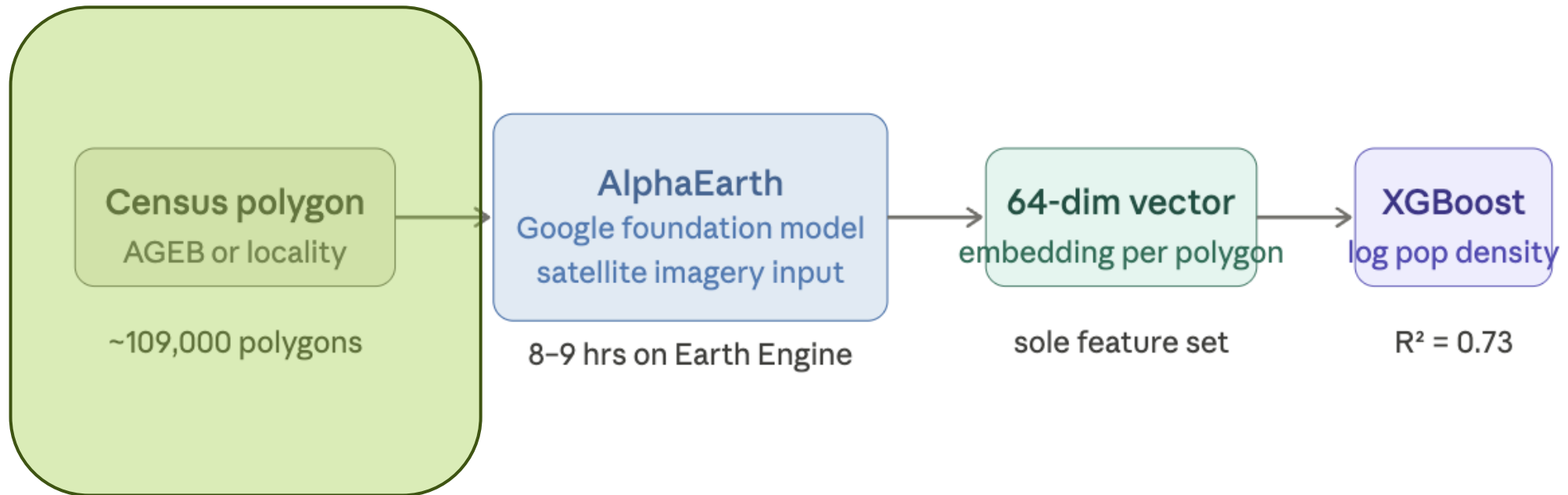
- Motivation
- Intervention
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- Results

My approach has four key stages:



Step 1: constructing census polygon

High-level project approach



I chose to study Mexico because it has great data availability and documentation.



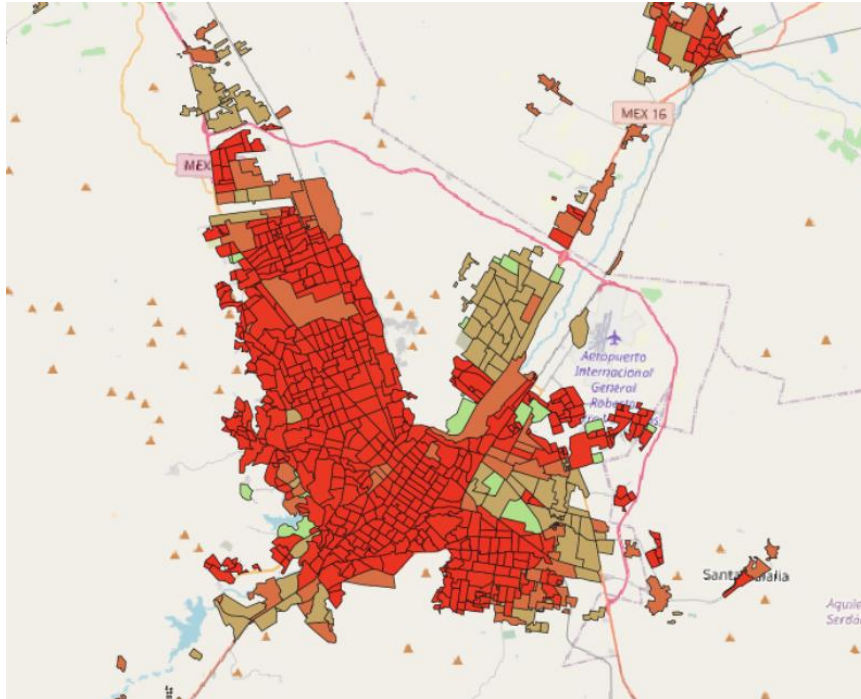
Data is available for censuses ranging from **1895–2020**

Population data reported at **locality** and **AGEB** levels

Locality represent human settlement.
It's any place inhabited with one or more dwellings that is known by a specific name (given by law or custom).

AGEBs are the fundamental geographic unit used for statistical data collection (similar to a **Census Tract** in the US).

Creating the census polygons required merging data across different data types and sources (1/2)

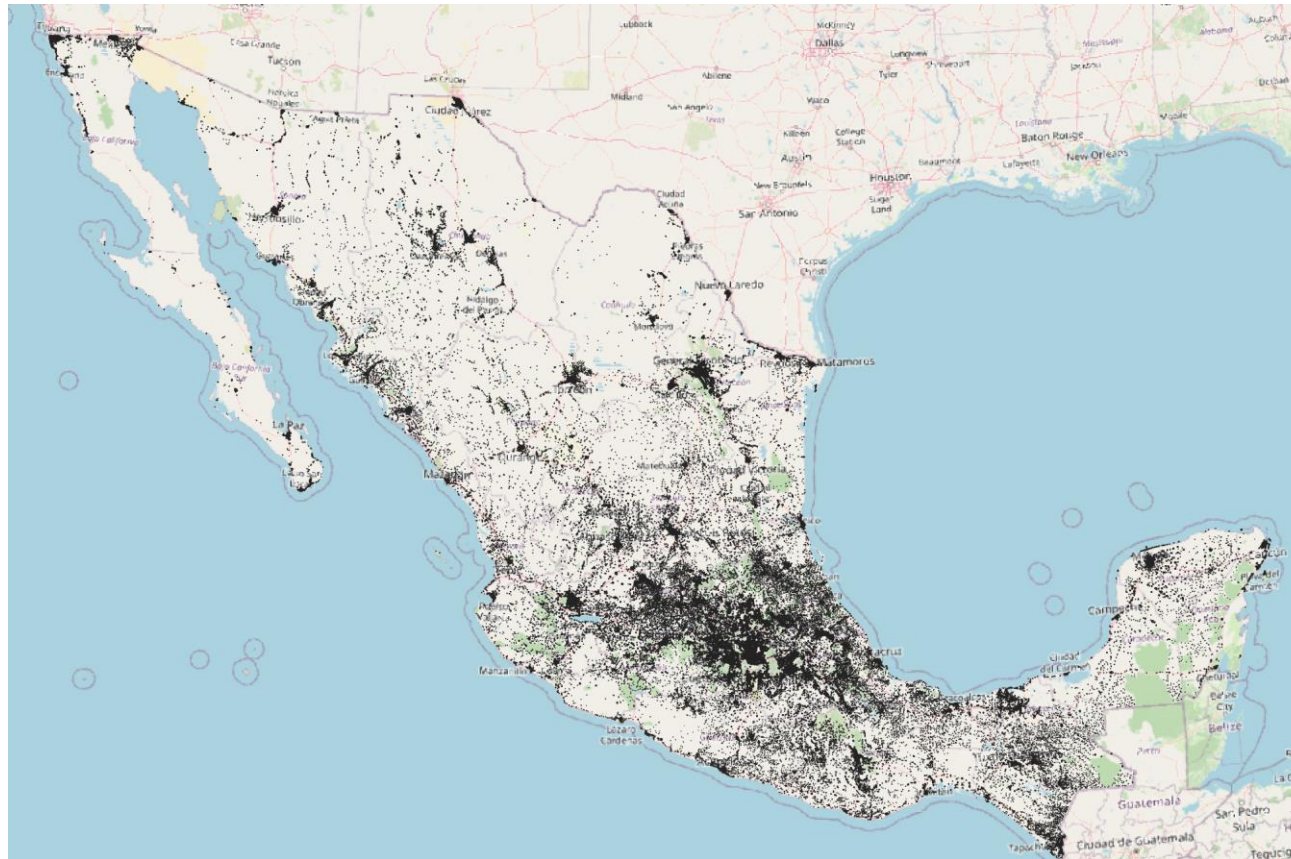


A	B	C	D	E	F	G	H	I	J	K	L	M	N	P
JAD	NOM_ENT	MUN	NOM_MUN	LOC	NOM_LOC	LONGITUD	LATITUD	ALTITUD	POBTOT	POBFEM	POBMAS	P_0A2	P_0A2_F	P_0
	Chihuahua	000	Total de la en	0000	Total de la Entidad				3741869	1888047	1853822	163950	80466	
	Chihuahua	000	Total de la en	9998	Localidades de una vivienda				17223	7361	9862	838	410	
	Chihuahua	000	Total de la en	9999	Localidades de dos viviendas				14611	5693	8918	728	356	
	Chihuahua	001	Ahumada	0000	Total del Municipio				14635	7258	7377	805	382	
	Chihuahua	001	Ahumada	0001	Miguel Ahumada	106°30'44.68 30°37'03.842		1197	10510	5303	5207	497	229	
	Chihuahua	001	Ahumada	0007	Álamos de Pe	106°47'44.35 30°18'13.067		1282	354	189	165	22	14	
	Chihuahua	001	Ahumada	0010	Argelia	105°38'11.65 30°24'56.888		1273	4 *	*	*	*	*	*
	Chihuahua	001	Ahumada	0011	Atotonilco	106°11'32.02 30°29'00.715		1395	5 *	*	*	*	*	*
	Chihuahua	001	Ahumada	0013	Barreal [Colc	106°15'01.63 30°07'39.706		1389	42	24	18	5	3	
	Chihuahua	001	Ahumada	0014	Buenos Aires	106°20'41.58 30°42'52.351		1228	19	10	9	1	0	
	Chihuahua	001	Ahumada	0020	Carrizal	106°39'10.60 30°33'41.463		1220	65	36	29	2	2	
	Chihuahua	001	Ahumada	0030	Las Nuevas C	105°54'28.97 30°03'09.574		1315	108	53	55	9	4	
	Chihuahua	001	Ahumada	0032	La Chiripa	105°59'15.03 30°34'55.590		1435	2 *	*	*	*	*	*
	Chihuahua	001	Ahumada	0050	Las Isabeles	106°45'38.92 30°16'09.514		1290	16	8	8	1	0	
	Chihuahua	001	Ahumada	0054	Leonor	106°31'27.95 30°48'33.842		1196	3 *	*	*	*	*	*
	Chihuahua	001	Ahumada	0055	San Fernando	106°27'10.35 30°49'39.977		1215	10 *	*	*	*	*	*
	Chihuahua	001	Ahumada	0058	Las Margarita	106°49'41.03 30°20'02.949		1288	4 *	*	*	*	*	*
	Chihuahua	001	Ahumada	0059	Medio Camir	106°26'54.61 30°10'12.356		1403	2 *	*	*	*	*	*
	Chihuahua	001	Ahumada	0064	Moctezuma	106°26'07.79 30°11'48.161		1384	50	22	28	1	0	
	Chihuahua	001	Ahumada	0068	El Nopal	106°03'29.96 30°00'22.378		1463	6 *	*	*	*	*	*
	Chihuahua	001	Ahumada	0069	Nueva Holan	106°46'41.03 30°16'24.076		1286	12	9	3	0	0	
	Chihuahua	001	Ahumada	0070	Ojo Caliente	106°32'59.36 30°29'37.761		1211	472	234	238	24	8	
	Chihuahua	001	Ahumada	0079	Las Playas	106°48'16.68 30°15'31.734		1293	202	85	117	11	3	
	Chihuahua	001	Ahumada	0085	Rancho Gran	106°46'02.60 30°18'49.054		1279	2 *	*	*	*	*	*
	Chihuahua	001	Ahumada	0086	Rancho Nuev	106°53'59.17 30°36'16.582		1214	48	21	27	0	0	
	Chihuahua	001	Ahumada	0097	San Luis	106°27'02.07 30°09'38.168		1411	17	6	11	1	0	
	Chihuahua	001	Ahumada	0110	Los Sauces	106°48'30.27 30°14'34.957		1298	3 *	*	*	*	*	*
	Chihuahua	001	Ahumada	0116	Las Tarabilla	106°14'34.91 29°57'20.915		1461	8 *	*	*	*	*	*
	Chihuahua	001	Ahumada	0119	El Toro (El Ág	105°36'07.33 30°22'49.013		1261	2 *	*	*	*	*	*
	Chihuahua	001	Ahumada	0121	Potrero del U	106°47'45.27 30°33'24.524		1220	24	7	17	1	0	
	Chihuahua	001	Ahumada	0122	Las Tunas	105°51'37.52 30°44'07.040		1410	14 *	*	*	*	*	*
	Chihuahua	001	Ahumada	0124	Matlanas [Cc	106°12'04.47 30°08'53.984		1406	82	33	49	12	6	

Shapefiles for the locality and AGEB distinctions in Mexico

Population data from census at AGEB and locality levels

Creating the census polygons required merging data across different data types and sources (2/2)



109,000 polygons!

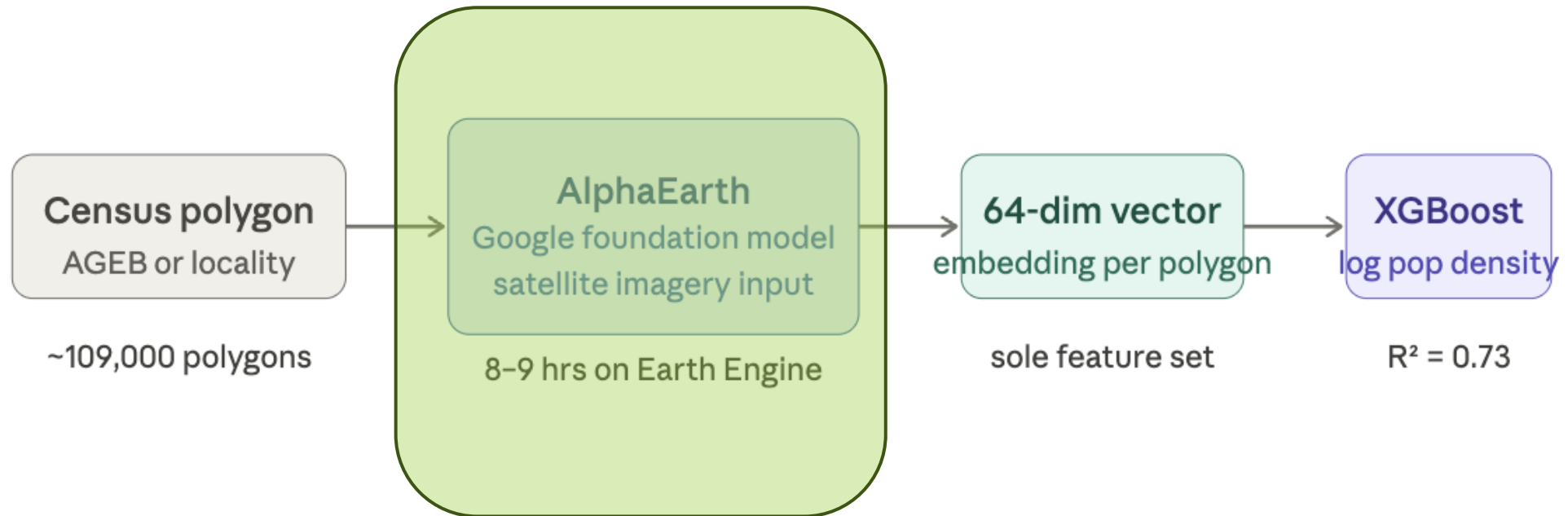
Black areas indicate polygons
or areas where I have data on
population

Empty areas indicate areas
with **no recorded population**

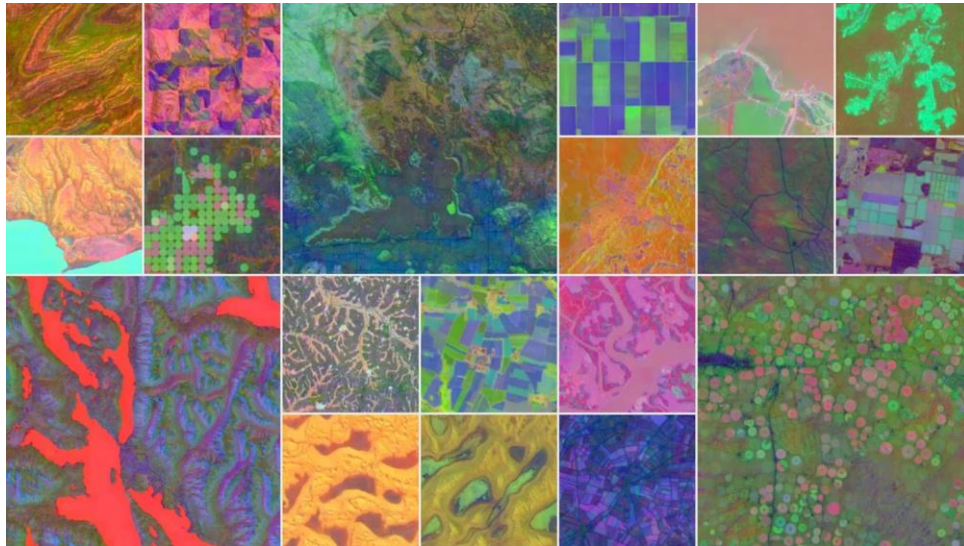
This is the input for next step
(AlphaEarth model)

Step 2: the Google AlphaEarth Foundation Model

High-level project approach



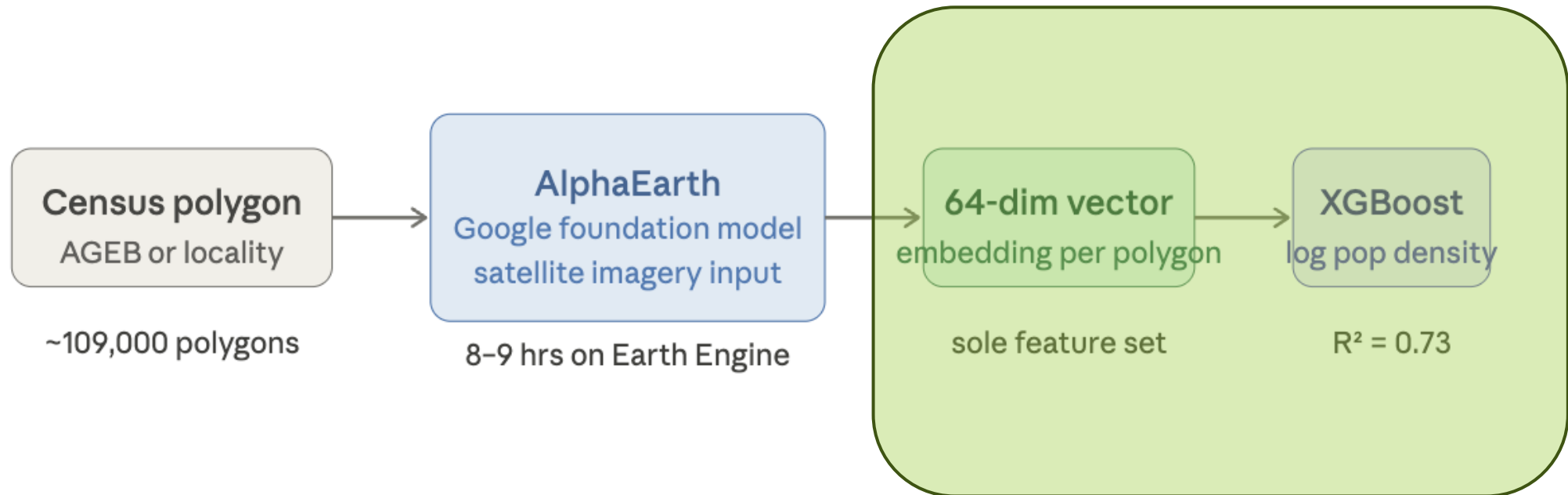
Google's AlphaEarth foundation model is a recently released state-of-the-art geospatial foundation model



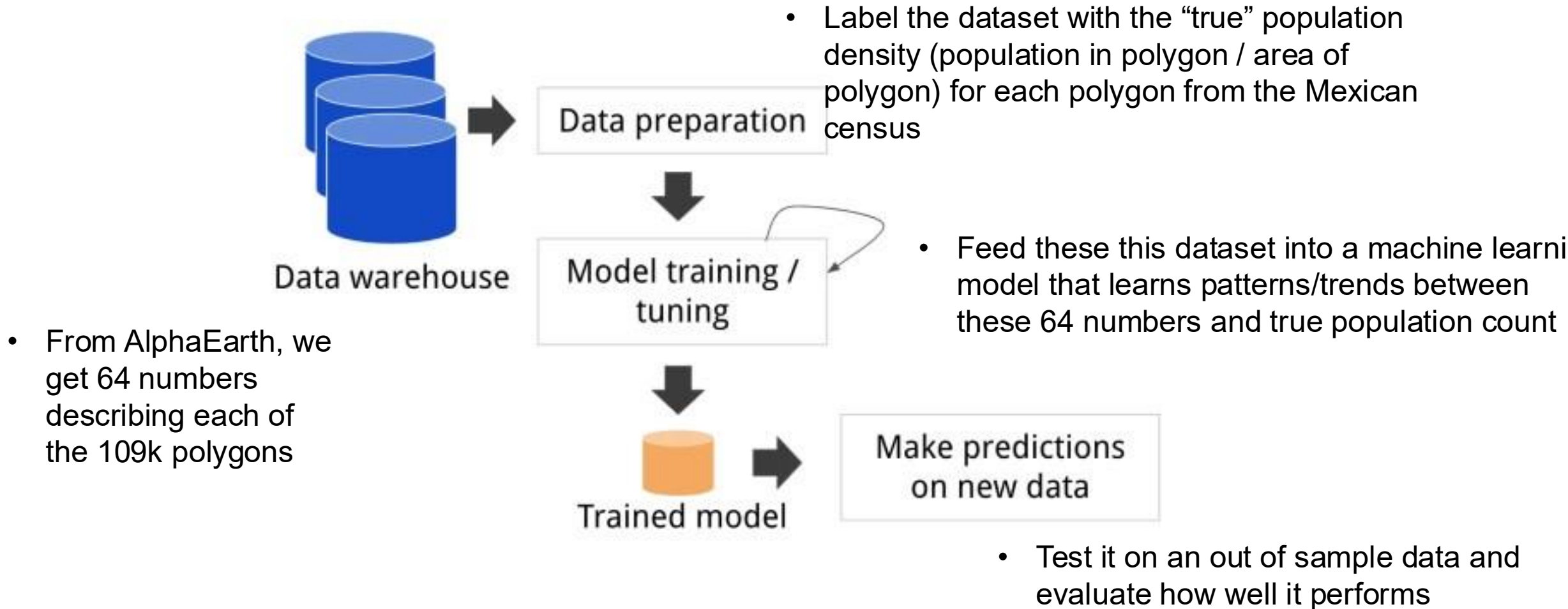
- Released in 2025
- Has **data from 2017– present day**
- **10m by 10m resolution.** For each 10m x 10m patch of earth's surface, it generates 64 number “embedding” that describes that area.
- Total: **96 trillion points of data.**
 - Earth's surface area = 150 million sqkm = 150 trillion sqm.
 - With 10m x 10m resolution, AlphaEarth distills Earth's surface area into 1.5 trillion patches of land.
 - For each patch, it assigns 64 numbers describing it

Step 3: Prediction

High-level project approach



Model specifics



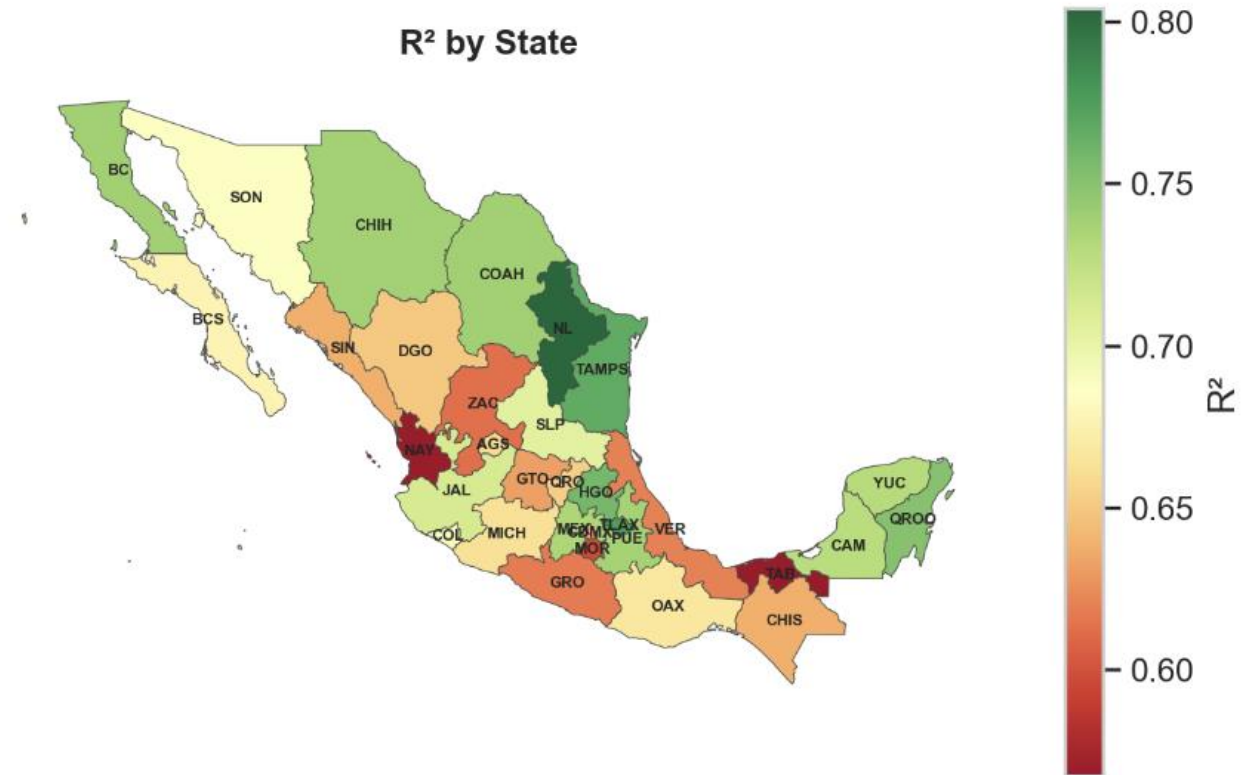
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- **Results**

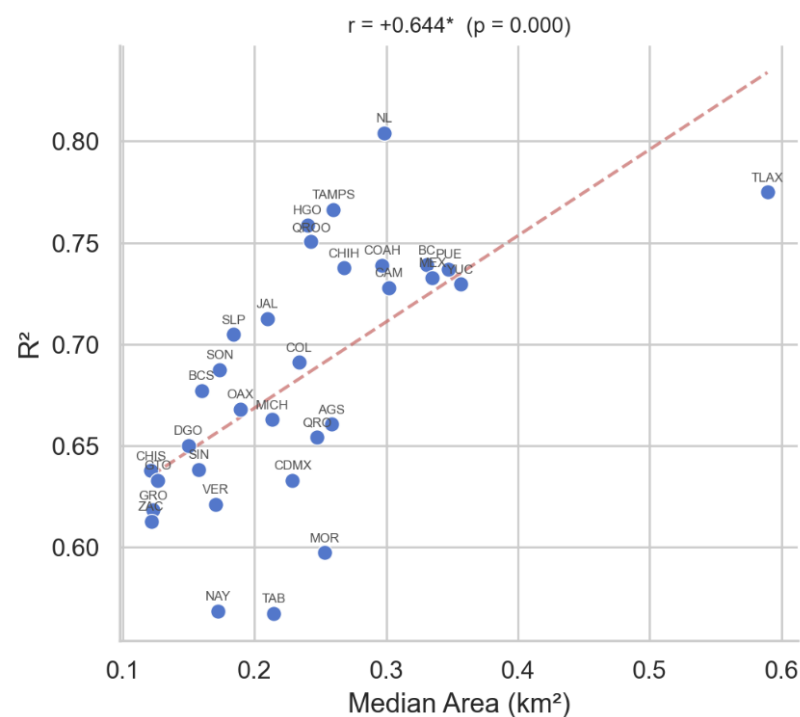
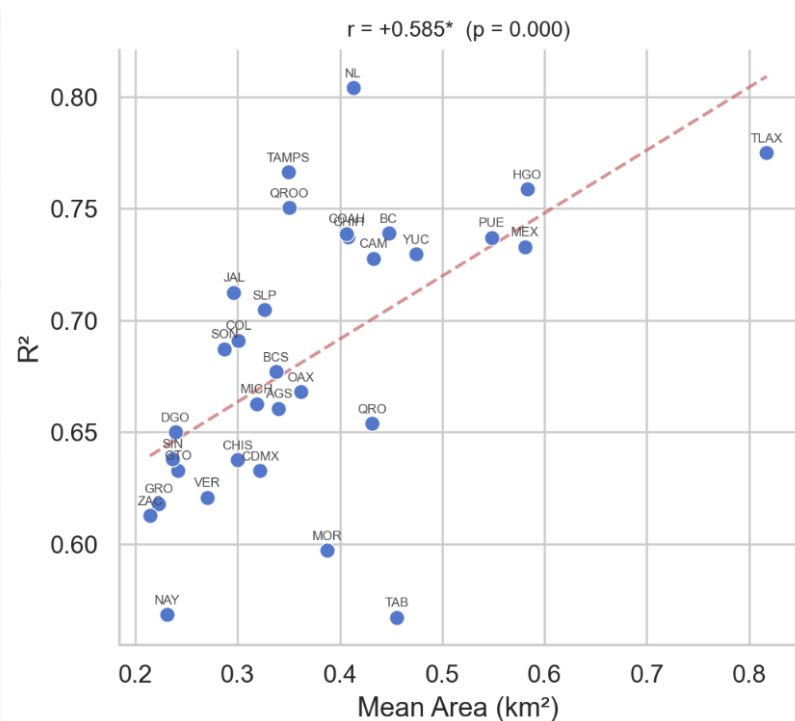
Key Results

Performance on par with existing task-specific models: 73% of the variability in population density can be explained by the AlphaEarth embeddings

Spatial differences in performance: Northern states tend to perform better than southern, more tropical states (?)



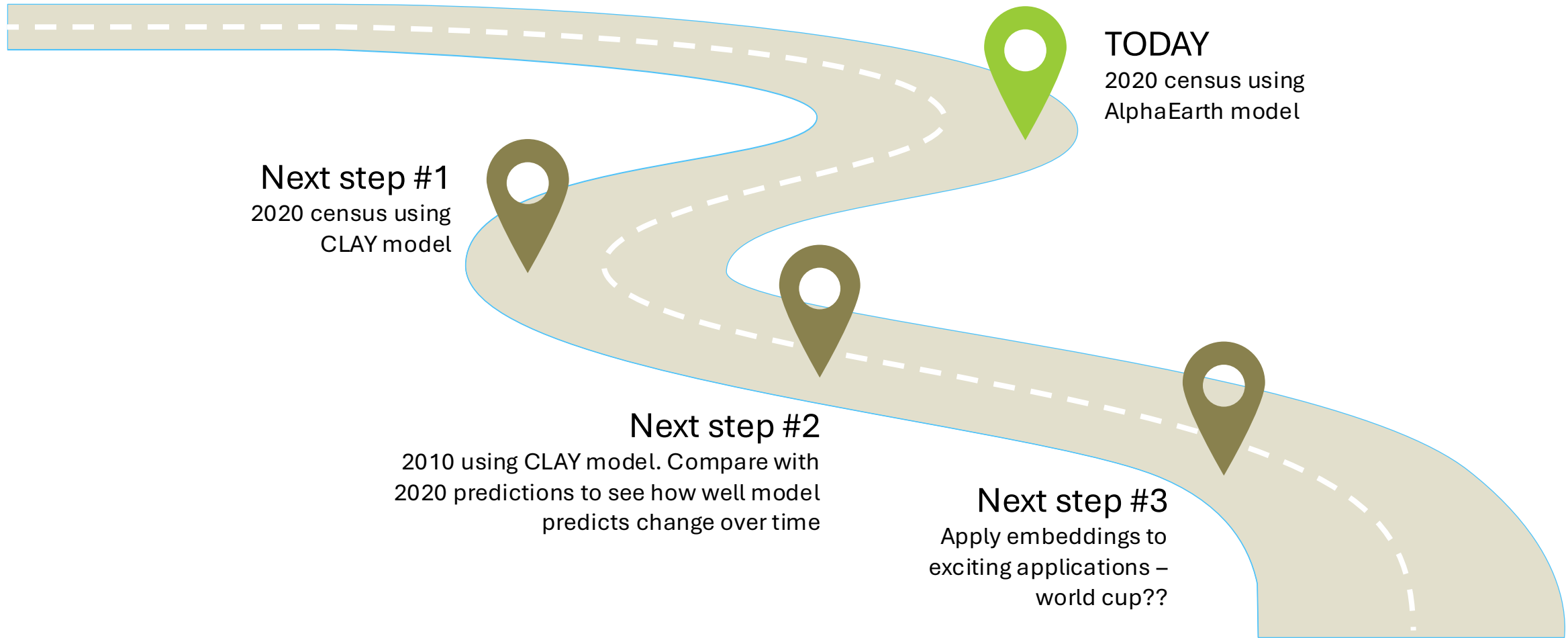
State-by-state performance variation driven by size of average polygon in the state



A larger polygon area correlates strongly with better R^2 model performance

Larger polygons average over more satellite pixels, which can produce stabler, less noisy embeddings

Next steps include expanding to other models and reviewing xyz.



Thank you! Questions?